3 **(a)** The density of fluid ρ, the cross-section area A of tube, the total mass M of tube and sand and g is constant, **B1**

hence acceleration a is proportional to displacement x from the equilibrium position. **B1**

The negative sign indicates that acceleration a is in opposite direction to displacement x from the equilibrium position. **B1**

These indicate that the motion of the tube is simple harmonic where $a = -\omega^2 x$. **A0**

(b) $$\omega^2 = \frac{\rho A g}{M}$$

$$= \frac{\left(1.2 \times 10^3\right)\left(5.3 \times 10^{-4}\right)9.81}{130 \times 10^{-3}}$$

$$= 48 \text{ rad}^2 \text{ s}^{-2}$$ **C1**

$$f = \frac{\omega}{2\pi} = 1.1 \text{ Hz}$$ **A1**

MC:	*There were some careless mistakes in converting cm^2 to SI units. In addition, some students mistakenly thought* $\omega = 48$ rad s^{-1}.

4 (a) The re-drawn Fig. 4.1 is as follows i.e. R, R and (R, R in series) are in